

AtlasReaper

Software Engineer / AI Infrastructure & Audio Systems

Dundee, United Kingdom
+44 (0) 7599 062287
atlas@atlas-systems.uk
atlas-systems.uk
linkedin.atlas-systems.uk
git.atlas-systems.uk

Profile

Final-year BA (Hons) Game Design and Production student at Abertay University and Junior Software Engineer at Simple Online Healthcare. Work spans production web engineering, automated testing, CI/release tooling, local AI infrastructure, and real-time audio systems. Selected through the 2026 Saltire Scholars Programme and retained part-time through final year, with a full-time pathway after graduation.

Founder of Atlas Systems, an independent engineering estate spanning local AI, automation, infrastructure, observability, retrieval systems, and interactive audio. Built SONIN, a dual-engine generative audio-visual system awarded 100% with faculty recognition for advanced DSP awareness and architectural maturity.

Experience

- Jun 2026–Present **Junior Software Engineer, Simple Online Healthcare, Remote**
- Contribute to production-facing Next.js/TypeScript healthcare journeys, including member-facing flows, supporting frontend implementation, bug fixing, API integration, and cross-platform debugging.
 - Expanded automated quality coverage using Vitest, React Testing Library, Playwright, and synthetic journey testing to improve confidence in member-facing flows and regressions.
 - Built reusable developer tooling and documented scenario-based development and testing patterns to make realistic local testing faster and more consistent.
 - Diagnosed and resolved CI/release failures across Changesets, GitHub Actions, and GitHub Packages, tracing issues through release metadata to verified publication.
 - Authored internal AI skills, guidance, and knowledge-base material to make AI-assisted development more consistent, safer, and easier to adopt across projects.
- Sept 2023–Present **Founder & Systems Engineer, Atlas Systems, Independent**
- Design and operate a governed technical estate spanning local AI, automation, infrastructure, observability, real-time audio, and 35 governed public repositories at atlas-systems.uk.
 - Built reusable GitHub Actions and Cloudflare delivery paths separating pull-request validation, isolated browser-evidence previews, production deployment, and post-deployment verification.
 - Built a local AI knowledge layer using FastAPI, Ollama, ChromaDB, type-aware chunking, hybrid vector/BM25 retrieval, cited answer synthesis, and explicit public/private source boundaries.
 - Developed interactive engineering systems including System Symphony, Spectral Forge, telemetry interfaces, architecture explorers, and observability surfaces backed by real estate data and deterministic evidence.
- Jun 2026–Present **Saltire Scholar, Entrepreneurial Scotland, Global**
- Selected through Scotland's national Saltire Scholars Programme. The placement at Simple Online Healthcare led to continued part-time employment through final year and a full-time pathway after graduation.
- Serve as a lead liaison between scholars and the programme team, helping coordinate communication, feedback, and engagement across the cohort.
 - Act as a Saltire brand ambassador, representing the programme and its values across professional, university, and external-facing settings.

Selected Engineering Projects

- 2026 **System Symphony, Atlas Systems**
- Built a read-only generative instrument that translates bounded infrastructure telemetry into deterministic, replayable, and auditable real-time music, with explicit treatment of unknown and unmeasured states.
- 2025 **SONIN: Dual-Engine Generative Audio-Visual System, Abertay University, Grade: 100%**
- The only cohort submission cited by faculty for “advanced DSP awareness” and “architectural maturity.”
- Designed a dual-engine Max/MSP architecture for real-time algorithmic audio and visual generation.
 - Engineered fault-tolerant data flow and compensation logic across independently operating systems.

Education

- 2023–2027 **BA (Hons) Game Design and Production**, *Abertay University*, Dundee
Industry-accredited games programme with a technical focus on interactive systems, audio implementation, production, and systems design.
○ **Final-year focus:** interactive systems, technical audio, AI-assisted workflows, software architecture, and production engineering.
- 2019–2021 **Aeronautical Engineering – Undergraduate Coursework**, *University of Glasgow*, Glasgow
Coursework included Thermodynamics, Fluid Mechanics, Engineering Mathematics, and Structural Analysis before transferring into games and interactive systems.
- 2013–2019 **Advanced Highers & Highers**, *Dollar Academy*
Advanced Highers: Mathematics (A), Physics (A), Engineering Science (A).

Technical Skills

- Languages Python, TypeScript/JavaScript, PHP, C++, C#
- Web React, Next.js, REST APIs, HTML/CSS
- AI / Retrieval Ollama, Open WebUI, ChromaDB, RAG, hybrid vector/BM25 retrieval
Local model evaluation, OpenAI and Anthropic API integration
- Testing Vitest, React Testing Library, Playwright, synthetic journey testing
- Platform GitHub Actions, Changesets, GitHub Packages, Docker
Cloudflare Workers, Pages, KV, D1, AWS, DNS/TLS, Git, Perforce
- Audio Unreal Engine 5, MetaSounds, Max/MSP, DSP, FL Studio, REAPER
- Production Agile/Scrum/Kanban, Jira, technical documentation

Recognition

- 2026 **Saltire Scholar**, *Entrepreneurial Scotland*
Selected through Scotland's national Saltire Scholars Programme.
- 2024 **Duke of Edinburgh's Gold Award**
Completed the Gold Award, including the expedition phase twice, serving as lead navigator and team mentor throughout.
- 2019 **Academic Excellence**, *SQA Advanced Highers*
Grade A across Mathematics, Physics, and Engineering Science.

Selected Technical Writing

W-08 | SPECULAR-CORE: Architectural Recovery

Architectural recovery of a live technical estate, documenting boundaries, failure modes, governance, and evidence.

W-06 | Atlas Systems CI/CD

Design and assurance of a controlled delivery pipeline with explicit validation, deployment, and verification stages.

W-03 | Ramone: Local AI System

Local AI architecture covering retrieval, reproducible infrastructure, grounded responses, and operational boundaries.

Earlier Experience

- May **F&B Supervisor**, *Crieff Hydro Hotel*, Crieff
- 2022–Apr 2023 Assumed full operational responsibility during an extended management absence and helped move guest satisfaction from the lowest-rated to the second-highest-rated department in the hotel during the same period.
- Jun 2021–Apr 2022 **Team Manager**, *Hunter Sales and Co.*, Glasgow
Promoted within the first quarter; built and led a B2B sales team, introducing structured conflict-resolution and planning processes that improved team output.